

KULDEEP SINGH

BMS, Charging & Vehicle Software Lead | Controls Validation and Firmware Integration

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PROFESSIONAL SUMMARY

EV software and electrification leader targeting vehicle software, BMS firmware, charging integration, and controls validation roles. Brings 8 years of safety-critical EV powertrain experience across eVCU/BMS application software, MATLAB/Simulink/Stateflow model-based controls, HV battery architecture, charging controller logic, DCDC/OBC/PDU integration, fault handling, CAN diagnostics, and vehicle-level validation. Delivered certified commercial EV retrofit systems and certified HV battery packs from 5 kWh to 300 kWh across LFP, NMC, LTO, and ultra-capacitor chemistries.

TARGET ROLE FIT

- Embedded Firmware Engineer - Battery Management System
- Software Integration Engineer - Power Conversion & Charging
- Vehicle Software Controls Validation Engineer
- Controls / Firmware Engineer - Charging & Energy Products

CORE STRENGTHS

- Built BMS and eVCU application software for EV systems where battery state, charger state, contactor/precharge conditions, voltage/current limits, and vehicle operating modes must work together safely.
- Led AC/DC charging controller logic and power conversion interfaces across charger, BMS, VCU/eVCU, contactors, DCDC, OBC, and PDU systems, supporting firmware behaviour for charging, power conversion, and highly physical vehicle products.
- Validated EV/hybrid systems through model checks, bench/vehicle testing, CAN diagnostics, fault analysis, calibration support, and certification documentation, with DFMEA/RCA and ISO 26262 functional-safety awareness.

PROFESSIONAL EXPERIENCE

Technical Manager - Product Development

IX Energy Pvt. Ltd., Noida, India | Jul 2018 - Present

- Developed BMS and eVCU software using MATLAB/Simulink/Stateflow MBD methodology, translating functional and safety requirements into control logic for drive readiness, charging mode, fault mode, and safe shutdown.
- Implemented charging controller and power conversion logic across AC/DC charging architectures, DCDC converter, OBC, PDU, HV battery, contactors, and vehicle safety interfaces.
- Designed HV battery systems from 5 kWh to 300 kWh across 72V-800V LFP, NMC, LTO, and ultra-capacitor architectures, combining software behaviour with pack design, BMU development, and BTMS/HVAC validation.
- Supported vehicle-level software integration and validation using CAN tools, diagnostics, issue debugging, calibration behaviour review, and structured problem solving.
- Managed cross-functional suppliers and internal teams across battery, motor, gearbox, DCDC, EPS, BCS, TCS, and HVAC to maintain component readiness and release-quality integration outputs.

SELECTED PROGRAMMES & PROOF

- LCV/LPT EV conversion: ARAI-homologated 6.5T EV with 80 kW drive and 53 kWh / 320V LFP battery; contributed eVCU/BMS logic, charging interfaces, DCDC/OBC/PDU coordination, diagnostics, and safe operating-state control.
- Certified battery packs: delivered 53 kWh/332V LFP electric truck pack, 11.5 kWh/332V LTO hybrid bus pack, 28 kWh NMC pack, and 0.5 kWh/400V ultra-capacitor hybrid bus pack.
- EV charging controller: led AC/DC charging architecture logic, HV-LV handshake, charger/BMS/VCU coordination, fault response, and vehicle safety readiness.

TECHNICAL TOOLKIT & CERTIFICATIONS

MATLAB, Simulink, Stateflow, Embedded C/C++, BMS, eVCU, HV battery, power conversion, charging firmware, AC/DC charging, DCDC, OBC, PDU, CAN, diagnostics, Vector CANoe, Peak CAN, ISO 26262, DFMEA, RCA, validation, calibration.

- Six Sigma Black Belt - Certified
- ISO 26262 Functional Safety for Road Vehicles - TUV SUD Certified
- IATF 16949:2016 Automotive Quality Management - TUV SUD Certified
- AWS IoT & SIMULINK - Amazon Web Services / MathWorks Certified

EDUCATION

B.Tech - Mechanical Engineering, VIT University, Vellore, Tamil Nadu | 2014 - 2018 | First Class, 75%